

DampWatch: Separating Physical Dissipation from Generalized-Alpha Integration Error

Computational Mechanics Projects
Reproducible technical research note

29 September 2026

Abstract

Time integration can remove mechanical energy even when the physical damping matrix is zero. DampWatch implements the generalized-alpha method for constant linear structural systems and audits physical dissipation, applied work, and the remaining discrete energy defect. An ideal two-mass system separates low and high frequencies, and an independent matrix-exponential reference measures state error. The recorded conservative setting exhibits relative energy drift 6.63×10^{-14} ; with high-frequency spectral radius 0.5, the low mode retains 99.9309% of its energy while the underresolved high mode is suppressed. Refinement recovers order 1.99835. The audit distinguishes energy conservation from phase accuracy and avoids interpreting a signed budget defect as a material parameter.

1 Introduction and mechanical system

High-frequency numerical damping can suppress unresolved structural modes without intentionally damping resolved motion. The generalized-alpha family [1] makes this tradeoff explicit. The research question is how physical dissipation and time-discretization effects can be reported separately without falsely equating stable energy behavior with accurate dynamics.

The unconstrained coordinates satisfy

$$M\ddot{u} + C\dot{u} + Ku = f(t), \quad u(0) = u_0, \quad \dot{u}(0) = v_0. \quad (1)$$

Matrices are real, constant, and symmetric; M is positive definite and C, K are positive semidefinite. Constraints must be eliminated before integration. Initial acceleration is obtained from this physical equation, not selected independently.

Using $x_{n+1-\alpha} = (1 - \alpha)x_{n+1} + \alpha x_n$, the discrete equilibrium is

$$Ma_{n+1-\alpha_m} + Cv_{n+1-\alpha_f} + Ku_{n+1-\alpha_f} = (1 - \alpha_f)f_{n+1} + \alpha_f f_n. \quad (2)$$

For prescribed asymptotic spectral radius $0 \leq \rho_\infty \leq 1$,

$$\alpha_m = \frac{2\rho_\infty - 1}{1 + \rho_\infty}, \quad \alpha_f = \frac{\rho_\infty}{1 + \rho_\infty}, \quad (3)$$

$$\gamma = \frac{1}{2} + \alpha_f - \alpha_m, \quad \beta = \frac{1}{4}(1 + \alpha_f - \alpha_m)^2. \quad (4)$$

These are old-state weights; references using complementary alpha conventions require conversion before numerical comparison. Loads are weighted endpoint samples, not a new evaluation at an intermediate time. Smooth linear problems have second-order accuracy, while high-frequency dissipation is controlled by ρ_∞ .

2 Solution algorithm and energy accounting

Predict $u^p = u_n + hv_n + h^2(1/2 - \beta)a_n$ and $v^p = v_n + h(1 - \gamma)a_n$. Solve for the new acceleration using

$$H = (1 - \alpha_m)M + (1 - \alpha_f)\gamma hC + (1 - \alpha_f)\beta h^2 K, \quad (5)$$

$$Ha_{n+1} = f_{n+1-\alpha_f} - \alpha_m Ma_n - C[(1 - \alpha_f)v^p + \alpha_f v_n] - K[(1 - \alpha_f)u^p + \alpha_f u_n]. \quad (6)$$

Then $u_{n+1} = u^p + \beta h^2 a_{n+1}$ and $v_{n+1} = v^p + \gamma h a_{n+1}$. A Cholesky factorization of H is reused for constant h . Python orchestrates the method with compiled NumPy/SciPy linear algebra. The full amplification map propagates independent basis states of $[u, hv, h^2 a]$, retaining the parasitic acceleration mode.

With arithmetic endpoint averages denoted by bars, report

$$E_n = \frac{1}{2}(v_n^T M v_n + u_n^T K u_n), \quad (7)$$

$$D_n = \sum_{j < n} h \bar{v}_j^T C \bar{v}_j, \quad W_n = \sum_{j < n} \bar{f}_j^T (u_{j+1} - u_j), \quad (8)$$

$$R_n = E_n - E_0 + D_n - W_n. \quad (9)$$

At $\rho_\infty = 1$, consistent initial acceleration yields the average-acceleration Newmark trajectory. Since $\Delta u = h\bar{v}$ and $\Delta v = h\bar{a}$, dotting mean equilibrium with Δu gives $R_n = 0$ up to roundoff. At smaller ρ_∞ , R_n includes integration and work/dissipation quadrature errors; it is neither a monotone algorithmic energy nor physical damping.

3 Experiment and validation

The model is $M = \text{diag}(1, 0.15)$ kg and $K = \begin{bmatrix} 604 & -600 \\ -600 & 600 \end{bmatrix}$ N/m. Natural frequencies are 1.86490 and 67.82715 rad/s. Initial modal energies are 0.50 and 0.18 J. The 12 s free study uses $h = 0.05$ s, deliberately underresolving the high mode. A forced study uses $C = 0.06M + 10^{-4}K$, $f = [0.8 \sin(1.2t), 0]^T$ N, $h = 0.02$ s, and $\rho_\infty = 1$.

Recorded check	Value
Undamped relative energy drift, $\rho_\infty = 1$	6.63×10^{-14}
Normalized forced budget defect, $\rho_\infty = 1$	7.53×10^{-14}
Low-mode energy retention, $\rho_\infty = 0.5$	0.9993095
High-frequency radius at $\omega h = 10^7$, $\rho_\infty = 0.5$	0.50001685
Finest observed state order	1.99835

An independently evaluated state-space matrix exponential [2] supplies the damped reference at 101 common times over 2 s. At $h = 0.0005$ s, the relative energy-weighted state error is 0.00595545. This quantifies accuracy rather than inferring it from stability or energy alone. Figure 1 visualizes selected recorded diagnostics.

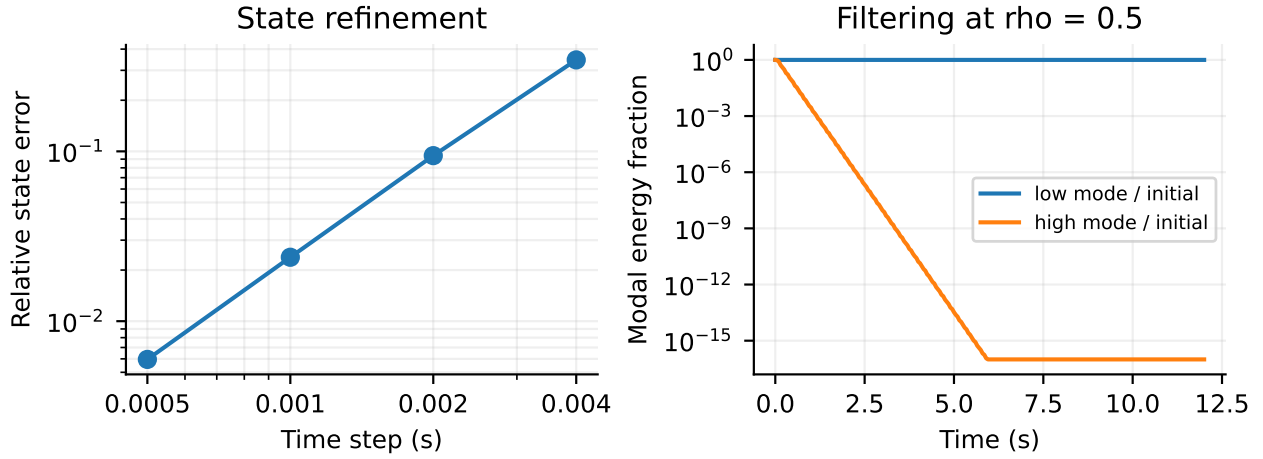


Figure 1: Recorded state convergence and modal-energy fractions with numerical filtering. The energy plot is clipped at 10^{-16} for readability; values below that level have no physical significance.

4 Discussion and limitations

Reducing ρ_∞ filters the poorly resolved attachment mode while preserving most low-mode energy. The high-mode final value below floating-point significance is not a physical measurement. The conservative parameter choice does not repair high-mode phase error. Likewise, D_n approximates physical dissipation along the numerical trajectory; it is not an exact continuous-time integral.

The solver handles dense, constant linear matrices only. It has no finite-element assembly, nonlinear/contact mechanics, time-dependent matrices, algebraic constraints, adaptive steps, or restart. Storage grows with the full time history, and dense factorization costs $O(n^3)$. Abrupt loads and extreme scaling require additional validation. With rigid modes the error metric is a seminorm; the showcase stiffness is positive definite. The matrix exponential is exact in time only up to floating-point and library error.

5 Reproducibility and conclusion

This note documents the committed implementation at `dc5a108ad5dc` and the existing synthetic results; it does not assert new solver runs. Install from the repository root with `python -m pip install ".[dev]"`; the README gives compiler requirements and verification commands. Code, data, and this manuscript are available at the project repository. No private course material is reproduced. The manuscript was prepared with AI assistance under the project owner's direction; no peer review is claimed. Run `dampwatch -output out`. The free response, forced ledger, spectral-radius, and convergence CSVs accompany `validation.json`. The key conclusion is that damping intent, discrete energy accounting, and state accuracy are separate questions. This combination of audits exposes the tradeoff more clearly than a displacement curve alone.

References

- [1] J. Chung and G. M. Hulbert, A time integration algorithm for structural dynamics with improved numerical dissipation: the generalized-alpha method, *J. Applied Mechanics* 60(2), 371–375 (1993). doi:10.1115/1.2900803.
- [2] SciPy, *scipy.linalg.expm* reference. Matrix-exponential documentation.